# ECE 276A Project 3: Visual-Inertial SLAM

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## I. Introduction

This project studies visual-inertial simultaneous localization and mapping (SLAM) using stereo visual features and inertial measurements. The goal is to estimate both the robot trajectory and a sparse landmark map from synchronized IMU velocities and stereo image observations. This problem is important because inertial dead reckoning alone accumulates drift over time, while visual observations provide geometric constraints that can be used to correct the pose estimate and improve long-term consistency.

My implementation follows the three required stages of the project. First, I estimate an IMU-only trajectory by propagating the pose with linear and angular velocity measurements. Second, I assume this trajectory is known and estimate 3D landmarks from stereo observations using triangulation and landmark-only EKF updates. Third, I combine the IMU prediction with visual landmark observations to perform visual-inertial SLAM. In the final implementation, I use the landmarks estimated in Question 3 as a fixed map and apply online pose correction in Question 4, which gives a more stable result than my earlier full joint update attempts.

## II. Problem Formulation

The goal of this project is to estimate the IMU trajectory and the 3D positions of visual landmarks from synchronized inertial and stereo-camera measurements. Let the IMU pose at time $t$ be

$$^{W}\mathbf{T}_{I,t} \in SE(3),$$

where $W$ denotes the world frame and $I$ denotes the IMU frame. Let the $j$-th landmark be

$$\mathbf{m}_j \in \mathbb{R}^3, \qquad j = 1, \dots, M.$$

The available measurements are:

- IMU linear velocity $\mathbf{v}_t \in \mathbb{R}^3$ and angular velocity $\boldsymbol{\omega}_t \in \mathbb{R}^3$, both expressed in the IMU frame,
- timestamps $\tau_t$,
- stereo visual feature observations

$$\mathbf{z}_{j,t} = \begin{bmatrix} u^l_{j,t} & v^l_{j,t} & u^r_{j,t} & v^r_{j,t} \end{bmatrix}^\top \in \mathbb{R}^4,$$

- left and right camera intrinsic matrices $\mathbf{K}_l, \mathbf{K}_r$,
- extrinsic calibration matrices

$$^{I}\mathbf{T}_L, {}^{I}\mathbf{T}_R \in SE(3),$$

  which map points from the regular left and right camera frames to the IMU frame.

The estimation problem is to recover

$$\mathcal{T} = \{^{W}\mathbf{T}_{I,t}\}_{t=1}^{T}, \qquad \mathcal{M} = \{\mathbf{m}_j\}_{j=1}^{M},$$

from the measurement history

$$\mathcal{Z}_{1:T} = \{\mathbf{z}_{j,t}\}, \qquad \mathcal{U}_{1:T} = \{(\mathbf{v}_t, \boldsymbol{\omega}_t)\}.$$

The IMU kinematic model is written in discrete time as

$$^{W}\mathbf{T}_{I,t+1} = {}^{W}\mathbf{T}_{I,t} \operatorname{Exp}\left( \begin{bmatrix} \mathbf{v}_t \\ \boldsymbol{\omega}_t \end{bmatrix} \Delta t_t \right), \qquad \Delta t_t = \tau_{t+1} - \tau_t.$$

For the visual observation model, the camera poses in the world frame are

$$^{W}\mathbf{T}_{L,t} = {}^{W}\mathbf{T}_{I,t}\, {}^{I}\mathbf{T}_L, \qquad {}^{W}\mathbf{T}_{R,t} = {}^{W}\mathbf{T}_{I,t}\, {}^{I}\mathbf{T}_R.$$

Since the provided extrinsics are for the regular camera frames, while the image projection model is defined in the optical frames, I additionally use a fixed regular-to-optical transform

$$^{O}\mathbf{T}_C = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Thus the predicted stereo observation of landmark $\mathbf{m}_j$ at time $t$ is

$$\hat{\mathbf{z}}_{j,t} = h\big({}^{W}\mathbf{T}_{I,t}, \mathbf{m}_j\big),$$

where $h(\cdot)$ denotes projection through the left and right optical cameras using $\mathbf{K}_l$ and $\mathbf{K}_r$.

The final objective is therefore:

1) estimate an IMU trajectory by propagating the $SE(3)$ pose with IMU velocities;
2) estimate landmark positions from stereo observations;
3) use the landmark observations to correct the IMU trajectory online through an EKF-based visual-inertial SLAM update.

## III. Technical Approach

### A. Preprocessing and Coordinate Handling

I first load the full feature tensor, the IMU measurements, the timestamps, and the calibration parameters. Since the feature tensor may appear in different layouts, I normalize it to the shape

$$\mathbf{Z} \in \mathbb{R}^{4 \times M \times T},$$

so that each feature observation is represented as

$$\mathbf{z}_{j,t} = \begin{bmatrix} u^l_{j,t} & v^l_{j,t} & u^r_{j,t} & v^r_{j,t} \end{bmatrix}^\top.$$

I also follow the corrected PR3 convention and treat the provided matrices `extL_T_imu` and `extR_T_imu` as

$$ {}^{I}\mathbf{T}_L, \qquad {}^{I}\mathbf{T}_R, $$

that is, transforms from the regular left/right camera frames to the IMU frame. The stereo projection is performed in the optical frames, so I apply the fixed regular-to-optical transform ${}^{O}\mathbf{T}_C$ before projection. This is consistent with my final code implementation. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8

### B. Question 1: IMU Localization via EKF Prediction

For Question 1, I estimate the IMU trajectory using only inertial motion propagation. Let

$$ \boldsymbol{\xi}_t = \begin{bmatrix} \mathbf{v}_t \\ \boldsymbol{\omega}_t \end{bmatrix} \in \mathbb{R}^6 $$

be the measured body-frame twist. The pose is propagated by

$$ {}^{W}\mathbf{T}_{I,t+1} = {}^{W}\mathbf{T}_{I,t} \operatorname{Exp}(\boldsymbol{\xi}_t \Delta t_t). $$

The corresponding covariance is propagated as

$$ \mathbf{P}_{t+1} = \mathbf{F}_t \mathbf{P}_t \mathbf{F}_t^\top + \mathbf{Q}_{\text{imu}} \Delta t_t^2, $$

where $\mathbf{F}_t$ is obtained from the matrix exponential of the negative adjoint twist matrix, and in my implementation

$$ \mathbf{Q}_{\text{imu}} = \operatorname{diag}(0.02, 0.02, 0.02, 0.005, 0.005, 0.005). $$

The initial covariance is set to

$$ \mathbf{P}_0 = 10^{-6}\mathbf{I}_6. $$

This step produces a smooth IMU-only baseline trajectory, but drift accumulates over time. :contentReference[oaicite:9]index=9

### C. Question 3: Landmark Mapping via EKF Update

For Question 3, I assume the predicted trajectory from Question 1 is known and focus only on the landmark state. Since the project statement assumes the landmarks are static, I do not use a prediction step for landmark motion and only perform EKF updates. :contentReference[oaicite:10]index=10

To control complexity, I first score feature tracks using:

- the total number of valid observations,
- the average stereo disparity magnitude.

I then keep all features satisfying

$$ \texttt{min_obs} = 6, $$

with `max_features=None`, so the full feature tensor is read and filtered holistically rather than truncating the feature list at load time. During each frame, I cap the number of processed observations by

$$ \texttt{max_updates_per_step} = 200. $$

The stereo measurement noise is

$$ \mathbf{R}_m = \sigma_m^2 \mathbf{I}_4, \qquad \sigma_m = 3.0 \text{ pixels}. $$

Each landmark is initialized by stereo triangulation:

$$ \mathbf{m}_j^{(0)} = \text{Triangulate}(\mathbf{z}_{j,t}, {}^{W}\mathbf{T}_{I,t}), $$

using two camera projection matrices

$$ \mathbf{P}_l = \mathbf{K}_l \, {}^{O}\mathbf{T}_C \, {}^{L}\mathbf{T}_W, \qquad \mathbf{P}_r = \mathbf{K}_r \, {}^{O}\mathbf{T}_C \, {}^{R}\mathbf{T}_W. $$

I solve the resulting DLT system

$$ \mathbf{A}\mathbf{X} = 0 $$

by SVD and convert the homogeneous solution to Euclidean world coordinates. After successful initialization, I set

$$ \boldsymbol{\Sigma}_j = 5\mathbf{I}_3, $$

while uninitialized landmarks start with

$$ \boldsymbol{\Sigma}_j = 100\mathbf{I}_3. $$

For an initialized landmark, the measurement model is

$$ \hat{\mathbf{z}}_{j,t} = h({}^{W}\mathbf{T}_{I,t}, \mathbf{m}_j). $$

The landmark Jacobian is computed numerically:

$$ \mathbf{H}_{m,j,t} = \frac{\partial h({}^{W}\mathbf{T}_{I,t}, \mathbf{m}_j)}{\partial \mathbf{m}_j}. $$

The EKF update is

$$ \mathbf{S}_{j,t} = \mathbf{H}_{m,j,t} \boldsymbol{\Sigma}_j \mathbf{H}_{m,j,t}^\top + \mathbf{R}_m, $$

$$ \mathbf{K}_{j,t} = \boldsymbol{\Sigma}_j \mathbf{H}_{m,j,t}^\top \mathbf{S}_{j,t}^{-1}, $$

$$ \mathbf{m}_j \leftarrow \mathbf{m}_j + \mathbf{K}_{j,t}(\mathbf{z}_{j,t} - \hat{\mathbf{z}}_{j,t}), $$

$$ \boldsymbol{\Sigma}_j \leftarrow (\mathbf{I} - \mathbf{K}_{j,t}\mathbf{H}_{m,j,t})\boldsymbol{\Sigma}_j(\mathbf{I} - \mathbf{K}_{j,t}\mathbf{H}_{m,j,t})^\top + \mathbf{K}_{j,t}\mathbf{R}_m\mathbf{K}_{j,t}^\top. $$

To improve robustness, I reject updates if:

- the raw reprojection residual satisfies

$$ \|\mathbf{z}_{j,t} - \hat{\mathbf{z}}_{j,t}\|_2 > 80, $$

- the Mahalanobis distance satisfies

$$ (\mathbf{z}_{j,t} - \hat{\mathbf{z}}_{j,t})^\top \mathbf{S}_{j,t}^{-1} (\mathbf{z}_{j,t} - \hat{\mathbf{z}}_{j,t}) > 25, $$

- the landmark norm becomes excessively large or non-finite.

I also clip each landmark increment to the interval $[-0.5, 0.5]$ componentwise. This produces a dense landmark map with some remaining outliers, but the dominant scene geometry is recovered well.

### D. Question 4: Visual-Inertial SLAM with Pose-Only Correction

For Question 4, I combine the IMU prediction with visual landmark observations to correct the IMU pose online. Instead of jointly optimizing pose and all landmarks online, I use the map estimated in Question 3 as a fixed map and update only the IMU pose. I chose this pose-only formulation because it is considerably more stable than my earlier full joint EKF version.

The prediction step is identical to Question 1:

$$
\begin{aligned}
{}^{W}\mathbf{T}_{I,t+1} &= {}^{W}\mathbf{T}_{I,t}\,\mathrm{Exp}(\boldsymbol{\xi}_t \Delta t_t),\\
\mathbf{P}_{t+1} &= \mathbf{F}_t \mathbf{P}_t \mathbf{F}_t^\top + \mathbf{Q}_{\text{pose}} \Delta t_t^2,
\end{aligned}
$$

with

$$
\mathbf{Q}_{\text{pose}} = \mathrm{diag}(0.03, 0.03, 0.03, 0.008, 0.008, 0.008).
$$

The stereo measurement noise is

$$
\mathbf{R}_p = \sigma_p^2 \mathbf{I}_4, \qquad \sigma_p = 2.5 \text{ pixels},
$$

and I again cap the number of processed observations by

$$
\texttt{max_updates_per_step} = 200.
$$

For each visible landmark from the fixed map, I compute the residual

$$
\mathbf{r}_{j,t} = \mathbf{z}_{j,t} - h({}^{W}\mathbf{T}_{I,t}, \mathbf{m}_j),
$$

and a numerical pose Jacobian

$$
\mathbf{H}_{p,j,t} = \frac{\partial h({}^{W}\mathbf{T}_{I,t}, \mathbf{m}_j)}{\partial \boldsymbol{\xi}}.
$$

The pose update is then

$$
\begin{aligned}
\mathbf{S}_{j,t} &= \mathbf{H}_{p,j,t} \mathbf{P}_t \mathbf{H}_{p,j,t}^\top + \mathbf{R}_p,\\
\mathbf{K}_{j,t} &= \mathbf{P}_t \mathbf{H}_{p,j,t}^\top \mathbf{S}_{j,t}^{-1},\\
\delta \boldsymbol{\xi}_{j,t} &= \mathbf{K}_{j,t} \mathbf{r}_{j,t}.
\end{aligned}
$$

I update the pose by right multiplication:

$$
{}^{W}\mathbf{T}_{I,t} \leftarrow {}^{W}\mathbf{T}_{I,t}\,\mathrm{Exp}(\delta \boldsymbol{\xi}_{j,t}).
$$

In practice, I apply several stabilization rules:

- reject observations with
$$
\|\mathbf{r}_{j,t}\|_2 > 80,
$$
- reject observations whose Mahalanobis distance exceeds 20,
- damp the pose increment by a factor of $0.25$,
- clip translation increments to $[-0.08, 0.08]$ and rotation increments to $[-0.02, 0.02]$.

After the pose update, the covariance is updated using the Joseph-form expression

$$
\mathbf{P}_t \leftarrow (\mathbf{I} - \mathbf{K}_{j,t}\mathbf{H}_{p,j,t})\mathbf{P}_t(\mathbf{I} - \mathbf{K}_{j,t}\mathbf{H}_{p,j,t})^\top + \mathbf{K}_{j,t}\mathbf{R}_p\mathbf{K}_{j,t}^\top.
$$

This design directly matches the current code and explains why the final SLAM result shows visible correction relative to the IMU-only baseline while still remaining stable. Compared with earlier versions, the corrected camera-geometry convention and optical-frame projection make the visual update significantly more effective.

### E. Visualization

For the final SLAM plots, I visualize the IMU trajectory and the landmark positions in the common world-frame $x$-$y$ plane. For readability only, I remove obviously far outlier landmarks in the final overlay figure using a norm threshold and a maximum distance-to-trajectory threshold. This filtering is applied only to the visualization and does not affect the SLAM estimate itself. The raw Question 3 landmark figures therefore still show a larger number of outliers than the final Question 4 overlay plots.

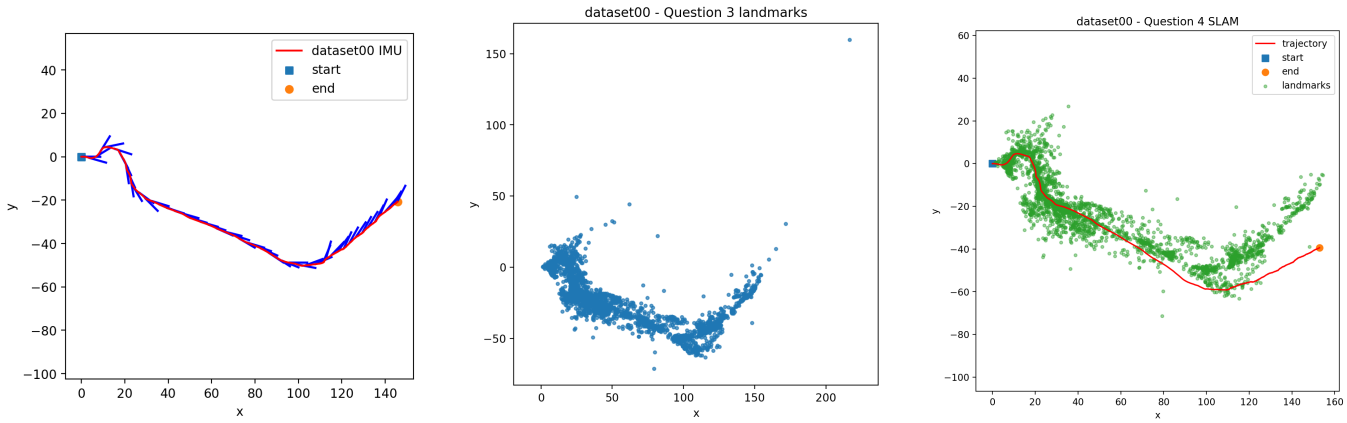
## IV. Results

Figures 1 and 2 show the IMU-only trajectory from Question 1, the landmark map from Question 3, and the final visual-inertial SLAM result from Question 4.

For **Dataset00**, the correction effect in Question 4 is now clearly visible. Compared with the IMU-only trajectory, the final SLAM trajectory shows a noticeable deviation in the second half of the path, which indicates that the landmark observations are actively correcting the inertial estimate. The dominant landmark structure in Question 3 follows the same overall U-shaped geometry as the trajectory, and the final SLAM result is more consistent with this structure than the IMU-only baseline. A residual offset still remains in the later segment. This is still reasonable, because Question 4 uses the fixed landmark map estimated in Question 3 rather than a jointly re-optimized map, and the pose update is deliberately conservative due to gating and damping. In addition, from the video sequence, many landmarks in the later segment are observed mainly on the left side of the robot path, so the landmark cloud is not expected to be symmetric around the trajectory.

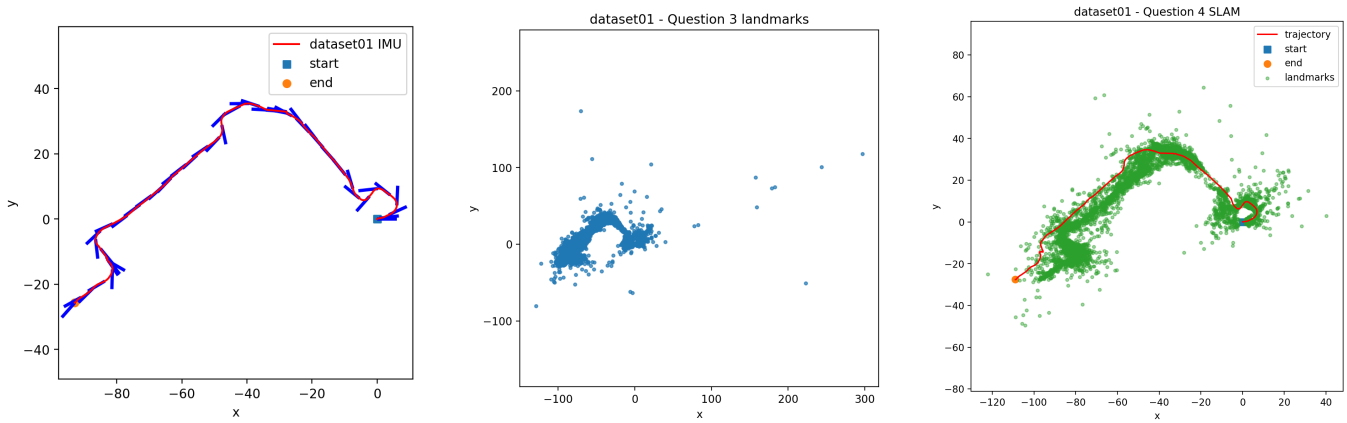
For **Dataset01**, the correction is especially clear near turning segments. The IMU-only trajectory contains several sharp heading changes, including an initial turn that appears close to a right-angle bend. After the visual-inertial update, this turn becomes smoother and less abrupt. Similar behavior also appears in later turns, where the SLAM trajectory has reduced local curvature compared with the IMU-only result. This suggests that the landmark observations are not only correcting translation drift, but also regularizing the heading estimate and the local shape of the path. The final trajectory remains consistent with the dominant landmark arc recovered in Question 3.

Overall, the final results are much more convincing than my earlier implementation. The correction in Question 4 is now clearly visible: in Dataset00 it appears mainly as a noticeable shift in the later part of the path, while in Dataset01 it appears mainly as smoother turning geometry. It is also important to note that the trajectory is not expected to pass through the center of the landmark cloud, since the landmarks correspond to static scene features rather than points on the robot path itself. Therefore, a one-sided or surrounding landmark distribution can still be physically reasonable. The remaining mismatch is mainly due to the use of a fixed landmark map together




(a) Dataset00 IMU-only trajectory from Question 1 (b) Dataset00 landmark map from Question 3 (c) Dataset00 final SLAM result from Question 4

Fig. 1. Results on Dataset00. The IMU-only trajectory provides a smooth baseline, Question 3 recovers the dominant landmark structure with several outliers, and Question 4 introduces visible landmark-based correction, especially in the later part of the trajectory, while improving consistency with the landmark map.




(a) Dataset01 IMU-only trajectory from Question 1 (b) Dataset01 landmark map from Question 3 (c) Dataset01 final SLAM result from Question 4

Fig. 2. Results on Dataset01. The final pose-only SLAM update keeps the trajectory stable, makes the turning geometry smoother than the IMU-only baseline, and improves consistency with the dominant landmark structure.

with a conservative pose-only update, but the overall geometric consistency is improved.